

**The Connectedness Among Financial Markets: Volatility  
Spillovers Before, During, and After the COVID-19 Pandemic**

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## Abstract

This paper investigates volatility spillovers among seven financial markets — the S&P 500, Bitcoin, gold, the CBOE Volatility Index (VIX), the S&P 500 futures index, the USD/EUR exchange rate, and US 5-year Treasury bonds — across three sub-periods: before, during, and after the COVID-19 pandemic (9 May 2016 to 6 November 2024). Volatility is estimated for each market using a GARCH(1,1) model, and pairwise Granger causality tests are applied to the resulting conditional volatility series in order to identify the direction and statistical significance of volatility transmission between markets, and the results are summarized in volatility spillover maps for each sub-period. Before the pandemic, cross-market linkages were present but comparatively weak, leaving meaningful scope for portfolio diversification. During the pandemic, connectedness intensified sharply: the S&P 500, its futures contract, and the VIX emerged as the dominant transmitters of volatility, propagating shocks to Bitcoin, gold, the exchange rate, and Treasury bonds. In the post-pandemic period, equity-related markets and the VIX remained significant sources of spillovers even as some bilateral links weakened. Taken together, the evidence supports the financial contagion hypothesis: precisely when diversification is most needed, cross-market correlations and volatility transmission rise, eroding its benefits. The findings carry practical implications for hedging strategies, risk management, and portfolio construction during periods of systemic stress.

*Keywords: financial markets; volatility spillover; Granger causality; connectedness; COVID-19; contagion.*

## 1. Introduction

Financial markets have become increasingly integrated over recent decades. Advances in trading technology, the globalization of capital flows, and the growing overlap of investor bases mean that a shock originating in one market can now propagate to others within days or even hours. Understanding how, and how strongly, volatility is transmitted across asset classes is therefore central to modern finance: it determines whether diversification genuinely reduces portfolio risk, how hedges should be constructed, and where regulators should look for the origins of systemic stress.

The COVID-19 pandemic provides an unusually clean natural experiment for studying these questions. It was an exogenous, global shock that hit all asset classes nearly simultaneously, and it separates the recent past into three economically distinct regimes: a comparatively calm pre-pandemic period, an acute crisis period, and a post-pandemic period characterized by inflation, monetary tightening, and the normalization of markets. If financial contagion is real, connectedness among markets should intensify during the crisis regime and the identity of the markets transmitting shocks should become clearly visible.

This paper maps volatility spillovers among seven markets spanning equities (S&P 500 and its futures contract), risk sentiment (VIX), cryptocurrency (Bitcoin), commodities (gold), foreign

exchange (USD/EUR), and fixed income (US 5-year Treasury bonds) over the period 9 May 2016 to 6 November 2024. Rather than focusing on a single market pair, the analysis covers the full cross-section of bilateral relationships, using pairwise Granger causality tests on GARCH-estimated volatility to establish both the existence and the direction of spillovers in each of the three sub-periods. The results are presented as volatility spillover maps — matrices of p-values in which each cell tests whether the volatility of one market helps predict the volatility of another.

Three main findings emerge. First, cross-market connectedness was present but modest before the pandemic, implying that diversification across stocks, gold, bonds, currencies, and Bitcoin was effective. Second, connectedness rose dramatically during the pandemic: the S&P 500, the S&P 500 futures index, and the VIX became near-universal transmitters of volatility, Granger-causing the volatility of virtually every other market in the system, while Treasury bond volatility also transmitted significantly to most markets. Third, in the post-pandemic period the equity complex and the VIX remained significant sources of spillovers, and volatility correlations across asset classes stayed well above their pre-pandemic levels, suggesting that the pandemic durably raised the degree of market integration. These results support the financial contagion hypothesis and imply that diversification benefits deteriorate exactly when investors need them most.

The remainder of the paper is organized as follows. Section 2 reviews the literature on volatility spillovers. Section 3 describes the data and methodology. Section 4 presents the empirical results, covering descriptive evidence, correlation analysis, and the Granger causality spillover maps. Section 5 discusses the economic interpretation and practical implications of the findings. Section 6 concludes and outlines directions for future research.

## **2. Literature Review**

### ***2.1 Methodological approaches***

A rich set of econometric methodologies has been developed to measure connectedness among financial markets. Generalized variance decomposition has been used to attribute forecast-error variance to shocks from other markets (BenSaïda et al., 2018; Bagheri and Ebrahimi, 2020; Lin and Chen, 2021), while the framework of Diebold and Yilmaz has become the workhorse for constructing aggregate and directional spillover indices (Liow, 2015; Qarni et al., 2019; Aslam et al., 2021). Time-varying parameter VAR (TVP-VAR) models allow connectedness to shift smoothly over time (Jiang et al., 2022; Bouri et al., 2022; Lin and Chen, 2021), and multivariate GARCH specifications — including BEKK (Li and Giles, 2014), EGARCH (Jebran and Iqbal, 2016), DCC-MGARCH (Khan, 2023), and heterogeneous spillover GARCH models (Wang et al., 2018; Mukherjee and Mishra, 2010) — capture second-moment transmission directly. Complementary approaches include Granger causality testing (Khan, 2023; Jeong and Park, 2018), network analysis (Lee and Lee, 2020; Jeong and Park, 2018), and regression-based designs (Bouri et al., 2022).

## ***2.2 Markets, assets, and regions studied***

The variables examined in this literature are similarly diverse: stock market indices and their volatilities (BenSaïda et al., 2018; Li and Giles, 2014; Wang et al., 2018; Amar et al., 2021; Aslam et al., 2021), public real estate (Liow, 2015), bonds (Qarni et al., 2019; Bagheri and Ebrahimi, 2020; Lee and Lee, 2020), exchange rates (Jiang et al., 2022; Udejaja, 2019), Bitcoin and other cryptocurrencies (Qarni et al., 2019; Bagheri and Ebrahimi, 2020; Jiang et al., 2022), crude oil and gold (Jiang et al., 2022; Naeem et al., 2021), futures contracts (Bagheri and Ebrahimi, 2020), and market turnover (Jeong and Park, 2018). Geographically, studies span Europe (Aslam et al., 2021; Bouri et al., 2022), Asia (Li and Giles, 2014; Mukherjee and Mishra, 2010; Jebran and Iqbal, 2016), the United States (Wang et al., 2018; Baruník et al., 2016), Latin America (Gamba-Santamaria et al., 2017; Amar et al., 2021), the BRICS economies (Khan, 2023), Korea (Jeong and Park, 2018), Northeast Asia (Lee and Lee, 2020), and Nigeria (Udejaja, 2019).

## ***2.3 Key findings and open questions***

Despite this methodological and geographical diversity, the literature converges on a consistent stylized fact: volatility spillovers among financial markets intensify during crises. BenSaïda et al. (2018), using a regime-switching framework, show that risk spillovers strengthen markedly in crisis regimes, with the US and Germany acting as transmitters and Hong Kong and Japan as receivers. Liow (2015) documents strong and shifting volatility links across G7 markets that peak during periods of stress, and Jeong and Park (2018) find the same pattern within the Korean financial sector. Aslam et al. (2021) extend this evidence to the COVID-19 period specifically, showing that European stock markets exhibited exceptionally high intraday volatility spillovers during the pandemic — demonstrating that spillovers surge not only during financial crises but during systemic risk episodes more broadly.

Spillovers are also not confined to equity markets. Qarni et al. (2019) and Jiang et al. (2022) document volatility transmission between Bitcoin and traditional financial markets, with Bitcoin typically acting as a transmitter and equity markets as receivers; Jiang et al. (2022) further conclude that Bitcoin functions as a hedge but not as a safe haven, a conclusion echoed for gold and bonds during COVID-19 by Naeem et al. (2021). Li and Giles (2014) show that spillovers operate interregionally as well as regionally, with the US market transmitting strongly and unidirectionally to Asian emerging markets, and Mukherjee and Mishra (2010) find that transmission is detectable even at the intraday frequency.

The direction of spillovers, however, remains contested. Mukherjee and Mishra (2010) conclude that spillovers within Asia flow predominantly one way, toward India, whereas Khan (2023) reaches the opposite conclusion for a later sample, finding that Indian stock market volatility spills over to the other BRICS markets. Similarly, while Lin and Chen (2021) identify the US stock market as the core global transmitter, Lee and Lee (2020) find only weak integration within Northeast Asia. These disagreements suggest that the direction and strength

of spillovers are regime- and period-dependent — precisely the question this paper addresses by comparing three clearly delineated sub-periods around the COVID-19 shock. In addition, relatively few studies test a cross-section of asset classes as broad as the one considered here (equities, futures, implied volatility, cryptocurrency, gold, foreign exchange, and government bonds) within a single unified design.

**Table 1. Summary of the literature on volatility spillovers**

| Author(s)               | Year | Variables                                                  | Methodology          | Key findings                                                                                                   | Period    |
|-------------------------|------|------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------|-----------|
| BenSaïda et al.         | 2018 | Volatility Index                                           | GVD                  | Spillovers stronger during crises; US and Germany transmit, HK and Japan receive.                              | 2001–2017 |
| Liow                    | 2015 | Stocks, Bonds, FX, Real Estate                             | Diebold–Yilmaz       | Strong and shifting volatility links across G7 markets.                                                        | 1997–2013 |
| Aslam et al.            | 2021 | Stock Market Index                                         | Diebold–Yilmaz       | Very high spillover during COVID-19.                                                                           | 2019–2020 |
| Qarni et al.            | 2019 | Bitcoin, Asset Prices                                      | Diebold–Yilmaz       | Low and short-term volatility spillovers between US Bitcoin and financial markets.                             | 2010–2018 |
| Jiang et al.            | 2022 | Bitcoin, Crude Oil, Commodities, FX                        | TVP-VAR              | Bitcoin transmits volatility but is not a safe haven.                                                          | 2001–2018 |
| Li and Giles            | 2014 | US and Asian Stock Market Indices                          | BEKK-GARCH           | The US market has strong unidirectional volatility spillovers to Asian emerging markets.                       | 1993–2012 |
| Mukherjee and Mishra    | 2010 | India and Asian Market Indices                             | GARCH                | Strong intraday return and volatility spillovers from major Asian markets to India.                            | 1997–2008 |
| Jebran and Iqbal        | 2016 | Asian Stock Markets                                        | EGARCH               | Asymmetric volatility spillovers between stock and forex markets in Asia.                                      | 1999–2014 |
| Khan                    | 2023 | BRICS Stock Market Index                                   | VAR, DCC-MGARCH      | Indian markets transmit volatility to BRICS.                                                                   | 2013–2021 |
| Wang et al.             | 2017 | US and Developed Market Indices                            | GARCH-class model    | US stock volatility strongly spills over to international markets.                                             | 1991–2015 |
| Gamba-Santamaria et al. | 2017 | LATAM and US Stock Market Indices                          | Diebold–Yilmaz       | The US and Brazil mostly transmit volatility.                                                                  | 2003–2016 |
| Barunik et al.          | 2016 | US Sector Index                                            | Diebold–Yilmaz       | Asymmetric volatility spillovers across US sectors.                                                            | 2005–2012 |
| Naeem et al.            | 2021 | Global Stock Index, Oil, Gold, Bitcoin                     | Diebold–Yilmaz       | Cryptocurrency, bonds, and gold are hedges but not safe havens during COVID-19.                                | 2013–2020 |
| Bagheri and Ebrahimi    | 2020 | Global Stock Indices, Bonds, Commodities, Cryptocurrencies | Diebold–Yilmaz, HVAR | Markets are highly connected, especially during crises.                                                        | 2006–2018 |
| Amar et al.             | 2021 | Global MSCI Indices                                        | Diebold–Yilmaz       | Strong interdependence among markets; Europe and North America as net transmitters.                            | 2012–2020 |
| Bouri et al.            | 2022 | Global Stock Indices, Sentiment Indices                    | TVP-VAR, QQ          | Investor happiness significantly affects return and volatility connectedness.                                  | 2008–2019 |
| Lee and Lee             | 2020 | Northeast Asia and US Market Indices                       | Diebold–Yilmaz       | Weak regional integration.                                                                                     | 2006–2016 |
| Jeong and Park          | 2018 | Korean Financial Market Indices                            | Network Analysis     | Connectedness in the Korean financial sector rises during crises.                                              | 1990–2015 |
| Lin and Chen            | 2021 | China and US Financial Market Indices                      | TVP-SV-VAR           | China and US financial markets are highly connected; the US stock market is the core transmitter.              | 2004–2019 |
| Udeaja                  | 2019 | Nigerian Stock Market Indices                              | Diebold–Yilmaz       | Connectedness among Nigerian financial markets is highly time-varying and stronger during domestic turbulence. | 2000–2018 |

*Note:* GVD = generalized variance decomposition; TVP-VAR = time-varying parameter vector autoregression; QQ = quantile-on-quantile; HVAR = heterogeneous VAR.

### 3. Data and Methodology

#### 3.1 Data

The analysis employs daily data on seven financial market variables: the S&P 500 index, Bitcoin, gold, the CBOE Volatility Index (VIX), the S&P 500 futures index, the USD/EUR exchange rate, and the US 5-year Treasury bond. These series jointly span the major asset classes through which shocks are most likely to propagate: cash equities, equity derivatives, implied equity risk, cryptocurrency, precious metals, foreign exchange, and sovereign fixed income.

The sample covers 9 May 2016 to 6 November 2024 and is partitioned into three sub-periods around the COVID-19 pandemic: a pre-pandemic period (9 May 2016 to 19 January 2020), a pandemic period (20 January 2020 to 30 June 2021), and a post-pandemic period (1 July 2021 to 6 November 2024). The pandemic window opens with the first confirmed COVID-19 case in the United States and closes with the broad normalization of economic activity by mid-2021; the post-pandemic window captures the subsequent inflationary episode, the monetary tightening cycle, and the return to more ordinary market conditions. Splitting the sample in this way allows the connectedness structure of the system to be compared across three economically distinct regimes.

### ***3.2 Returns and volatility***

For each series, daily returns are computed from closing prices, and volatility is estimated using a GARCH(1,1) model. In this specification, the conditional variance of returns evolves as a function of the previous period's squared innovation and the previous period's conditional variance, capturing the volatility clustering that is characteristic of financial data — the tendency of turbulent days to follow turbulent days. The fitted conditional volatility series for each market serves as the input to the spillover analysis, with the VIX entering directly as a model-free measure of implied equity volatility. Working with volatility rather than returns focuses the analysis on the transmission of risk: two markets may exhibit low return correlation while their levels of turbulence nonetheless move together, which is the phenomenon of interest for hedging and risk management.

### ***3.3 Granger causality and the spillover map***

The direction of volatility transmission is identified using pairwise Granger causality tests. A variable  $X$  is said to Granger-cause a variable  $Y$  if past values of  $X$  contain information that improves the prediction of  $Y$  beyond the information already contained in past values of  $Y$  itself. For each ordered pair of markets and each sub-period, the null hypothesis that the volatility of the source market does not Granger-cause the volatility of the receiving market is tested; rejection of the null ( $p < 0.05$ ) is interpreted as a statistically significant volatility spillover from the source to the receiver.

The full set of pairwise tests is then arranged into a volatility spillover map: a matrix in which rows denote transmitting markets, columns denote receiving markets, and each cell reports the p-value of the corresponding test. Reading along a row reveals how widely a market broadcasts its volatility to the rest of the system; reading down a column reveals how exposed a market is

to volatility originating elsewhere. Because the tests are directional, the map distinguishes transmitters from receivers — information that symmetric correlation measures cannot provide. Granger causality is well suited to this design: it is transparent, imposes minimal structure on the data, and delivers an easily interpretable directional result for every market pair, which is exactly what is required to construct a system-wide map of spillovers.

## 4. Empirical Results

### 4.1 Correlation analysis

Tables 2–4 report the correlation matrices of the seven markets in levels for each sub-period, and Tables 5–6 report the correlations among the volatility series before and after the pandemic. Before COVID-19 (Table 3), cross-market relationships were weak: Bitcoin was essentially uncorrelated with the S&P 500 ( $-0.001$ ), gold co-moved only mildly with bonds ( $0.364$ ) and the exchange rate ( $-0.354$ ), and the only strong relationships were the mechanical ones — the near-perfect link between the S&P 500 and its futures contract, and the familiar strong negative relationship between the S&P 500 and the VIX ( $-0.793$ ).

During the pandemic (Table 2), the structure changed markedly. Bitcoin's correlation with the S&P 500 jumped from effectively zero to  $0.390$ , gold's correlation with equities turned positive ( $0.297$ ), and Treasury bonds became more strongly negatively correlated with equities ( $-0.448$ ). In other words, assets that had previously behaved independently began moving together, and the scope for diversification narrowed accordingly. After the pandemic (Table 4), several of these elevated co-movements persisted: Bitcoin's correlation with the S&P 500 remained at  $0.385$ , and the USD/EUR exchange rate became meaningfully correlated with equities ( $-0.335$ ), consistent with the dollar's role during the 2022–2023 tightening cycle.

The volatility correlations tell an even starker story. Before the pandemic (Table 5), volatility co-movement across asset classes was modest — for instance, S&P 500 volatility correlated at only  $0.135$  with gold volatility and  $0.227$  with bond volatility. After the pandemic (Table 6), every pairwise volatility correlation among the traditional assets increased substantially: S&P 500 volatility correlated at  $0.492$  with gold volatility,  $0.524$  with bond volatility,  $0.652$  with exchange-rate volatility, and  $0.439$  with Bitcoin volatility. Turbulence in one market is now far more likely to coincide with turbulence in the others than it was before 2020, indicating that the pandemic produced a durable increase in the integration of risk across markets.

**Table 2. Correlation matrix of financial assets during the COVID-19 period (20 Jan 2020 – 30 Jun 2021)**

| Variables   | (1)    | (2)    | (3)    | (4)   | (5) | (6) | (7) |
|-------------|--------|--------|--------|-------|-----|-----|-----|
| (1) S&P 500 | 1.000  |        |        |       |     |     |     |
| (2) Bitcoin | 0.390  | 1.000  |        |       |     |     |     |
| (3) VIX     | -0.703 | -0.335 | 1.000  |       |     |     |     |
| (4) Gold    | 0.297  | 0.260  | -0.210 | 1.000 |     |     |     |

|                     |        |        |        |        |        |        |       |
|---------------------|--------|--------|--------|--------|--------|--------|-------|
| (5) S&P 500 Futures | 1.000  | 0.390  | -0.703 | 0.297  | 1.000  |        |       |
| (6) USD/EUR         | -0.044 | -0.078 | 0.046  | -0.216 | -0.044 | 1.000  |       |
| (7) US 5Y T-Bonds   | -0.448 | -0.067 | 0.249  | 0.080  | -0.448 | -0.123 | 1.000 |

Note: (1) S&P 500; (2) Bitcoin; (3) VIX; (4) Gold; (5) S&P 500 futures index; (6) USD/EUR; (7) US 5-year Treasury bonds.

**Table 3. Correlation matrix of financial assets before the COVID-19 period (9 May 2016 – 19 Jan 2020)**

| Variables           | (1)    | (2)    | (3)    | (4)    | (5)    | (6)    | (7)   |
|---------------------|--------|--------|--------|--------|--------|--------|-------|
| (1) S&P 500         | 1.000  |        |        |        |        |        |       |
| (2) Bitcoin         | -0.001 | 1.000  |        |        |        |        |       |
| (3) VIX             | -0.793 | -0.021 | 1.000  |        |        |        |       |
| (4) Gold            | -0.042 | 0.054  | 0.019  | 1.000  |        |        |       |
| (5) S&P 500 Futures | 1.000  | -0.001 | -0.793 | -0.042 | 1.000  |        |       |
| (6) USD/EUR         | -0.040 | -0.018 | 0.038  | -0.354 | -0.040 | 1.000  |       |
| (7) US 5Y T-Bonds   | -0.343 | 0.058  | 0.251  | 0.364  | -0.343 | -0.120 | 1.000 |

Note: Variable numbering as in Table 2.

**Table 4. Correlation matrix of financial assets after the COVID-19 period (1 Jul 2021 – 6 Nov 2024)**

| Variables           | (1)    | (2)    | (3)    | (4)    | (5)    | (6)    | (7)   |
|---------------------|--------|--------|--------|--------|--------|--------|-------|
| (1) S&P 500         | 1.000  |        |        |        |        |        |       |
| (2) Bitcoin         | 0.385  | 1.000  |        |        |        |        |       |
| (3) VIX             | -0.729 | -0.342 | 1.000  |        |        |        |       |
| (4) Gold            | 0.257  | 0.136  | -0.193 | 1.000  |        |        |       |
| (5) S&P 500 Futures | 1.000  | 0.385  | -0.729 | 0.257  | 1.000  |        |       |
| (6) USD/EUR         | -0.335 | -0.189 | 0.221  | -0.352 | -0.335 | 1.000  |       |
| (7) US 5Y T-Bonds   | 0.065  | 0.037  | 0.022  | 0.298  | 0.065  | -0.288 | 1.000 |

Note: Variable numbering as in Table 2.

**Table 5. Correlation matrix of asset volatilities before the COVID-19 period**

| Variables                    | (1)   | (2)    | (3)    | (4)   | (5)   | (6)   |
|------------------------------|-------|--------|--------|-------|-------|-------|
| (1) Volatility S&P 500       | 1.000 |        |        |       |       |       |
| (2) Volatility Gold          | 0.135 | 1.000  |        |       |       |       |
| (3) Volatility US 5Y T-Bonds | 0.227 | 0.286  | 1.000  |       |       |       |
| (4) Volatility Bitcoin       | 0.157 | -0.437 | -0.128 | 1.000 |       |       |
| (5) Volatility USD/EUR       | 0.150 | 0.253  | 0.291  | 0.067 | 1.000 |       |
| (6) VIX                      | 0.740 | 0.158  | 0.101  | 0.099 | 0.045 | 1.000 |

Note: (1) Volatility of the S&P 500; (2) volatility of gold; (3) volatility of US 5-year Treasury bonds; (4) volatility of Bitcoin; (5) volatility of USD/EUR; (6) VIX.

**Table 6. Correlation matrix of asset volatilities after the COVID-19 period**

| Variables              | (1)   | (2) | (3) | (4) | (5) | (6) |
|------------------------|-------|-----|-----|-----|-----|-----|
| (1) Volatility S&P 500 | 1.000 |     |     |     |     |     |



|                              |       |       |       |       |       |       |
|------------------------------|-------|-------|-------|-------|-------|-------|
| (2) Volatility Gold          | 0.492 | 1.000 |       |       |       |       |
| (3) Volatility US 5Y T-Bonds | 0.524 | 0.362 | 1.000 |       |       |       |
| (4) Volatility Bitcoin       | 0.439 | 0.259 | 0.173 | 1.000 |       |       |
| (5) Volatility USD/EUR       | 0.652 | 0.469 | 0.672 | 0.101 | 1.000 |       |
| (6) VIX                      | 0.769 | 0.323 | 0.426 | 0.343 | 0.485 | 1.000 |

*Note: Variable numbering as in Table 5.*

## 4.2 Volatility spillover maps

Tables 7–9 present the volatility spillover maps for the three sub-periods. Each cell reports the p-value from a Granger causality test of the null hypothesis that the volatility of the row market does not Granger-cause the volatility of the column market; p-values below 0.05 indicate a statistically significant spillover from the row (transmitting) market to the column (receiving) market.

Before the pandemic (Table 8), significant spillovers were relatively sparse. The S&P 500 and its futures contract transmitted to the VIX and to each other, the VIX transmitted to the equity complex, and a handful of scattered links appear elsewhere — Bitcoin to Treasury bonds ( $p = 0.047$ ), the exchange rate to gold ( $p = 0.011$ ), and Treasury bonds to the futures index ( $p = 0.011$ ). Most bilateral relationships, however, are statistically insignificant: gold Granger-caused no other market, and Bitcoin, gold, and the exchange rate were largely disconnected from the core of the system. The pre-pandemic system was therefore only loosely connected, consistent with the weak correlations in Table 3.

During the pandemic (Table 7), the map transforms. The VIX Granger-caused the volatility of every other market in the system (all p-values 0.000), and the S&P 500 futures index did the same for every market except the VIX itself. The S&P 500 transmitted significantly to gold, the futures index, the exchange rate, and Treasury bonds, while Treasury bond volatility — reflecting the extraordinary stress in the Treasury market in March 2020 and the subsequent policy response — transmitted significantly to nearly all markets. Bitcoin and the exchange rate, by contrast, acted primarily as receivers, transmitting to only a small number of markets. The overall picture is one of dense, hierarchical connectedness: risk originated in the equity and implied-volatility complex and radiated outward to cryptocurrency, gold, currencies, and bonds.

After the pandemic (Table 9), connectedness recedes from its crisis peak but does not return to the pre-pandemic configuration. The VIX and the S&P 500 futures index remain broad transmitters, each Granger-causing five of the six other markets, and the S&P 500 continues to transmit to Bitcoin, the exchange rate, and (marginally) gold. Notably, Bitcoin, gold, the exchange rate, and Treasury bonds transmit little in this period — gold's only significant spillover is to Bitcoin ( $p = 0.038$ ) — so the post-pandemic system is characterized by a clear one-way hierarchy in which the equity complex and implied volatility drive the risk of the other asset classes.

**Table 7. Volatility spillover map during the COVID-19 period**

| To → / From ↓              | S&P 500 | Bitcoin | VIX   | Gold  | S&P 500 Futures | USD/EUR | US 5Y T-Bonds |
|----------------------------|---------|---------|-------|-------|-----------------|---------|---------------|
| <b>S&amp;P 500</b>         | X       | 0.113   | 0.290 | 0.003 | 0.000           | 0.000   | 0.000         |
| <b>Bitcoin</b>             | 0.708   | X       | 0.396 | 0.054 | 0.850           | 0.002   | 0.000         |
| <b>VIX</b>                 | 0.000   | 0.000   | X     | 0.000 | 0.000           | 0.000   | 0.000         |
| <b>Gold</b>                | 0.470   | 0.196   | 0.782 | X     | 0.058           | 0.000   | 0.286         |
| <b>S&amp;P 500 Futures</b> | 0.000   | 0.000   | 0.513 | 0.000 | X               | 0.000   | 0.000         |
| <b>USD/EUR</b>             | 0.334   | 0.416   | 0.753 | 0.073 | 0.056           | X       | 0.195         |
| <b>US 5Y T-Bonds</b>       | 0.020   | 0.052   | 0.032 | 0.000 | 0.002           | 0.000   | X             |

*Note: Cells report p-values from pairwise Granger causality tests of the null hypothesis that the volatility of the row market does not Granger-cause the volatility of the column market. p-values below 0.05 indicate a significant volatility spillover from the row market to the column market.*

**Table 8. Volatility spillover map before the COVID-19 period**

| To → / From ↓              | S&P 500 | Bitcoin | VIX   | Gold  | S&P 500 Futures | USD/EUR | US 5Y T-Bonds |
|----------------------------|---------|---------|-------|-------|-----------------|---------|---------------|
| <b>S&amp;P 500</b>         | X       | 0.195   | 0.000 | 0.123 | 0.705           | 0.389   | 0.777         |
| <b>Bitcoin</b>             | 0.791   | X       | 0.598 | 0.343 | 0.357           | 0.537   | 0.047         |
| <b>VIX</b>                 | 0.000   | 0.783   | X     | 0.456 | 0.000           | 0.750   | 0.445         |
| <b>Gold</b>                | 0.668   | 0.209   | 0.768 | X     | 0.757           | 0.683   | 0.622         |
| <b>S&amp;P 500 Futures</b> | 0.000   | 0.210   | 0.103 | 0.677 | X               | 0.236   | 0.038         |
| <b>USD/EUR</b>             | 0.378   | 0.956   | 0.455 | 0.011 | 0.130           | X       | 0.229         |
| <b>US 5Y T-Bonds</b>       | 0.059   | 0.063   | 0.127 | 0.543 | 0.011           | 0.059   | X             |

*Note: See note to Table 7.*

**Table 9. Volatility spillover map after the COVID-19 period**

| To → / From ↓              | S&P 500 | Bitcoin | VIX   | Gold  | S&P 500 Futures | USD/EUR | US 5Y T-Bonds |
|----------------------------|---------|---------|-------|-------|-----------------|---------|---------------|
| <b>S&amp;P 500</b>         | X       | 0.018   | 0.720 | 0.051 | 0.940           | 0.003   | 0.143         |
| <b>Bitcoin</b>             | 0.149   | X       | 0.481 | 0.656 | 0.124           | 0.746   | 0.150         |
| <b>VIX</b>                 | 0.000   | 0.035   | X     | 0.015 | 0.000           | 0.005   | 0.021         |
| <b>Gold</b>                | 0.652   | 0.038   | 0.348 | X     | 0.708           | 0.409   | 0.154         |
| <b>S&amp;P 500 Futures</b> | 0.000   | 0.001   | 0.384 | 0.035 | X               | 0.000   | 0.015         |
| <b>USD/EUR</b>             | 0.959   | 0.319   | 0.441 | 0.201 | 0.667           | X       | 0.194         |
| <b>US 5Y T-Bonds</b>       | 0.820   | 0.070   | 0.742 | 0.526 | 0.486           | 0.656   | X             |

*Note: See note to Table 7.*

### 4.3 Synthesis

Combining the correlation and Granger causality evidence yields a coherent narrative. Before COVID-19, the seven markets formed a loosely connected system in which diversification across asset classes was genuinely effective. The pandemic triggered a regime of intense connectedness in which the S&P 500, its futures contract, and the VIX became the dominant sources of system-wide risk, transmitting volatility to Bitcoin, gold, foreign exchange, and bonds, with Treasury bond volatility acting as a secondary transmitter. After the pandemic, the density of significant links declined, but the hierarchical structure persisted — the equity complex and the VIX still drive the system — and volatility correlations across asset classes remained far above their pre-pandemic levels. The COVID-19 shock therefore did not merely cause a temporary surge in connectedness; it appears to have durably tightened the integration of risk across financial markets, consistent with the financial contagion hypothesis.

## **5. Discussion and Implications**

### ***5.1 Financial contagion***

The results align closely with the contagion literature. As in BenSaïda et al. (2018), Liow (2015), and Aslam et al. (2021), connectedness intensifies sharply during the crisis regime; and as in Lin and Chen (2021) and Wang et al. (2018), the US equity market sits at the center of the transmission network. The finding that Bitcoin received rather than transmitted volatility during the pandemic complements Jiang et al. (2022) and Naeem et al. (2021): an asset that becomes a receiver of equity-market shocks precisely during stress episodes may serve as a hedge in calm times but cannot function as a safe haven when it matters most. The same logic applies to gold, whose correlation with equities turned positive during the pandemic.

### ***5.2 Implications for investors***

For portfolio construction, the central implication is that diversification benefits are state-dependent. Correlation and spillover estimates from tranquil periods materially overstate the protection a diversified portfolio will provide in a crisis, because both correlations and volatility transmission rise exactly when large losses occur. Risk managers should therefore stress-test portfolios under crisis-regime connectedness rather than full-sample averages, and should treat equity-market and VIX dynamics as leading indicators of turbulence in ostensibly unrelated asset classes: the spillover maps show that shocks to the equity complex propagate to cryptocurrency, gold, currencies, and bonds with a lag, which is precisely the structure that makes monitoring transmitters valuable.

### ***5.3 Implications for policymakers***

For regulators and central banks, the identification of persistent transmitters matters for systemic risk monitoring. If the equity complex and implied volatility drive risk across asset classes even in normal times — as the post-pandemic map suggests — then surveillance focused on equity and derivatives markets provides early warning for stress in bond, currency, and commodity markets. The strong transmission from Treasury bond volatility during the pandemic also underlines how quickly stress in the world's benchmark safe asset can radiate through the entire system, supporting the case for the market-functioning interventions deployed in March 2020.

### ***5.4 Limitations***

Several limitations should be acknowledged. Pairwise Granger causality identifies predictive relationships between two markets at a time and does not control for the influence of the remaining markets; a multivariate system could attribute spillovers more precisely. Granger causality is also a statement about predictability rather than structural causation, and the binary significant/insignificant reading of the maps does not quantify the magnitude of each spillover. Finally, the sub-period boundaries, while economically motivated, are necessarily somewhat

arbitrary, and results at the daily frequency may understate the intraday transmission documented elsewhere in the literature (Mukherjee and Mishra, 2010; Aslam et al., 2021).

## **6. Conclusion**

This paper mapped volatility spillovers among seven major financial markets before, during, and after the COVID-19 pandemic using pairwise Granger causality tests on daily volatility series. The evidence shows that financial markets are interconnected, and that the strength and structure of that interconnection is regime-dependent. Before the pandemic the system was loosely linked and diversification across asset classes was effective. During the pandemic, the S&P 500, the S&P 500 futures index, and the VIX became near-universal transmitters of volatility, propagating shocks to Bitcoin, gold, the USD/EUR exchange rate, and Treasury bonds, while cross-asset volatility correlations rose sharply. After the pandemic, the equity complex and the VIX remained the dominant drivers of system-wide risk, and volatility correlations stayed well above pre-pandemic levels, indicating a durable increase in market integration.

These results support the financial contagion hypothesis and carry a practical warning: during systemic risk episodes, it becomes substantially harder for investors to protect capital through diversification alone. Future research should extend this analysis with multivariate frameworks — such as VAR-based Diebold–Yilmaz spillover indices or TVP-VAR connectedness measures — to quantify the magnitude of each spillover, control for common shocks, and trace the evolution of connectedness continuously over time rather than across discrete sub-periods.

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